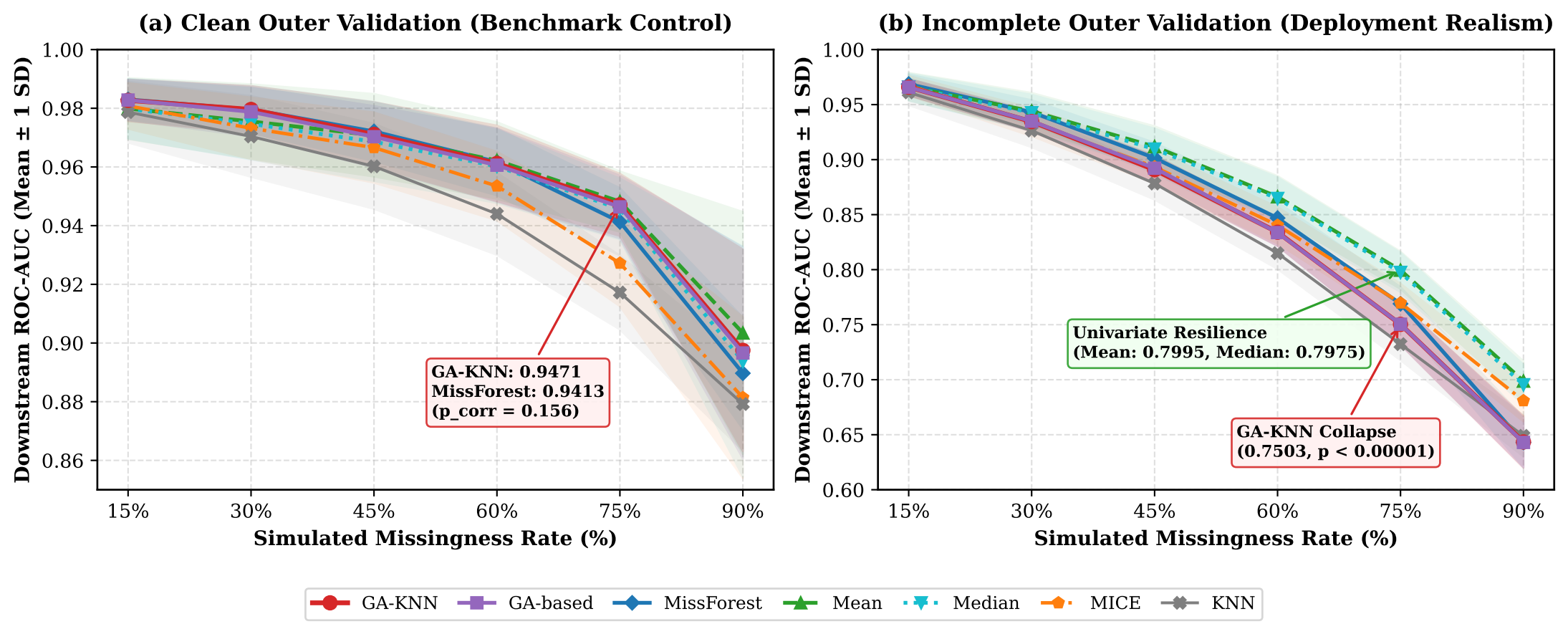
(a) Clean Outer Validation (Benchmark Control)
(b) Incomplete Outer Validation (Deployment Realism)
Downstream ROC-AUC (Mean ± 1 SD)
Simulated Missingness Rate (%)
15%
30%
45%
60%
75%
90%
GA-KNN: 0.9471
MissForest: 0.9413
(p_corr = 0.156)
Univariate Resilience
(Mean: 0.7995, Median: 0.7975)
GA-KNN Collapse
(0.7503, p < 0.00001)
GA-KNN
GA-based
MissForest
Mean
Median
MICE
KNN